

COMP30026

Models of Computation

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Lecture 16 – Proving Nonregularity via Fooling Sets



Hello!

- I work on Theoretical Computer Science
 - Algorithms for optimization problems with rigorous guarantees
- 4th time teaching COMP30026
- Last year, overhauled content on proving nonregularity and improved student outcomes and satisfaction
- <https://williamumboh.com/>

Challenging Material Ahead



- Budget extra time for:
 - Multiple passes over content
 - Practice on many problems
- Strongly recommended:
 - Tutorial attendance
 - Student hub support sessions
 - Consultation hours
 - Only 1 student attending so far!
- Significant difficulty increase in last 3 weeks
 - If currently behind, recommend prioritizing this lecture
 - Only requires basic knowledge of finite automata (Lecture 11)

Administrivia

- Lecture slides do not have all necessary details
 - Arguments too involved to include all details on slides
- Website with detailed notes
 - Online later today
 - Mobile-friendly
 - Search function
 - Students loved it last year 😊

Roadmap

- Last lecture: DFA cannot recognize $\{0^n 1^n \mid n \geq 0\}$ because it is **memoryless**
- This lecture: **Fooling set technique** to prove a language is not regular
- First impossibility result!

What does it mean to prove nonregularity?

Recall

1. Language is *regular* if and only if some DFA recognizes it
2. DFA M recognizes language L :

For every string x , M accepts x if and only if $x \in L$

To show L is *not regular*, need to show no DFA recognizes it, i.e.

For **every** DFA M , there exists a string x on which M makes a mistake:

- M accepts x but $x \notin L$; or
- M rejects x but $x \in L$

More Concretely

- Consider language $L = \{0^n 1^n \mid n \geq 0\}$ and the following DFA M

Strings	in L ?	Accepted by M ?
ϵ	Yes	Yes
0	No	No
1	No	Yes
00	Yes	Yes
$\vdots$	$\vdots$	$\vdots$

More Concretely

- Consider language $L = \{0^n 1^n \mid n \geq 0\}$ and the following DFA M

Strings	In L ?	Accepted by M ?
ϵ	Yes	Yes
0	No	No
1	No	Yes
00	Yes	Yes
$\vdots$	$\vdots$	$\vdots$

- M does not recognize L since it makes a mistake on 1
- To show L not regular, need to show every DFA makes a mistake
- Challenge:** Infinitely many DFAs!

Roadmap: Fooling Sets

1. Formalize **memorylessness** property of DFAs

Helpful to show that M makes a mistake

2. Lower bound on #states to recognize L , e.g.

“no DFA with fewer than 5 states can recognize L ”

3. To prove L not regular, show

“For every $k \geq 1$, no DFA with fewer than k states can recognize L ”

Extended Transition Function

- Let M be a DFA with state transition function δ and initial state q_0
- Recall: Let q be current state and a be an input symbol
 $\delta(q, a)$ = state after processing a from current state q
- The *extended transition function* δ^* of M is as follows: Let w be a string
 $\delta^*(q, w)$ = state after processing w from current state q
- The state $\delta^*(q_0, w)$ is the *resulting state* M on w and also written as $q(w)$

Warmup: 1-State DFAs

Lemma: A 1-state DFA either accepts all strings or accepts none.

Proof: For every string w , its resulting state $q(w) = q_0$.

If q_0 is accepting, then it accepts all strings

Else, it accepts none.

QED

Consequently: Every language that is not empty or all strings cannot be recognized by 1-state DFAs.

Prefix-Suffix Decomposition

Given a string w , can represent it as $w = xy$ where

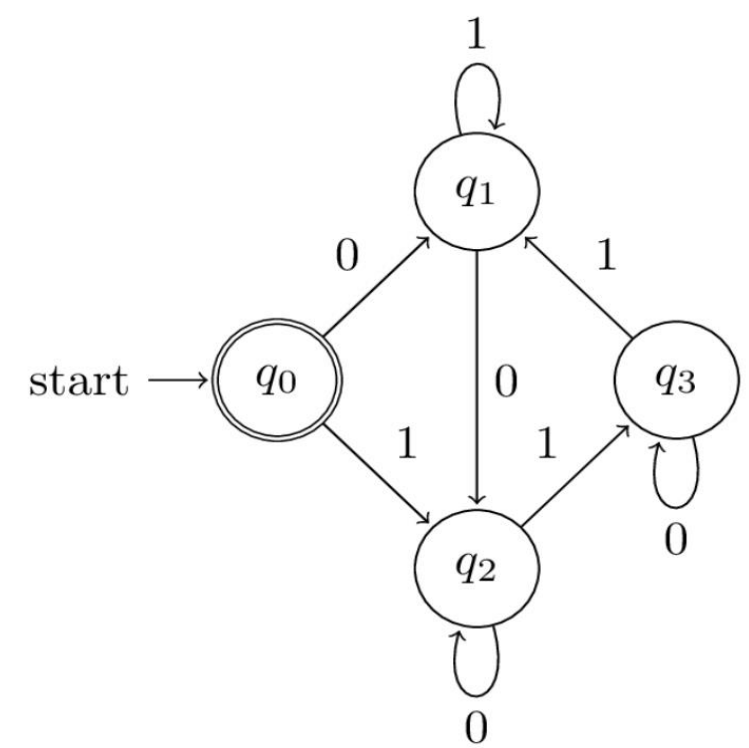
- x is a prefix (possibly empty) and
- y is a suffix (possibly empty)

Examples:

- $0011 = 0011$
- $0011 = 0011$
- $0011 = 0011$

Prefix-Suffix Decomposition

- Let M be the following DFA



Computation paths of M :

- On string **0****1****0****1****1**: $q_0 \xrightarrow{0} q_1 \xrightarrow{1} q_1 \xrightarrow{0} q_2 \xrightarrow{1} q_3 \xrightarrow{1} q_1$
- On string **0****0****1****1**: $q_0 \xrightarrow{0} q_1 \xrightarrow{0} q_2 \xrightarrow{1} q_3 \xrightarrow{1} q_1$
- On string **1****1****1**: $q_0 \xrightarrow{1} q_2 \xrightarrow{1} q_3 \xrightarrow{1} q_1$

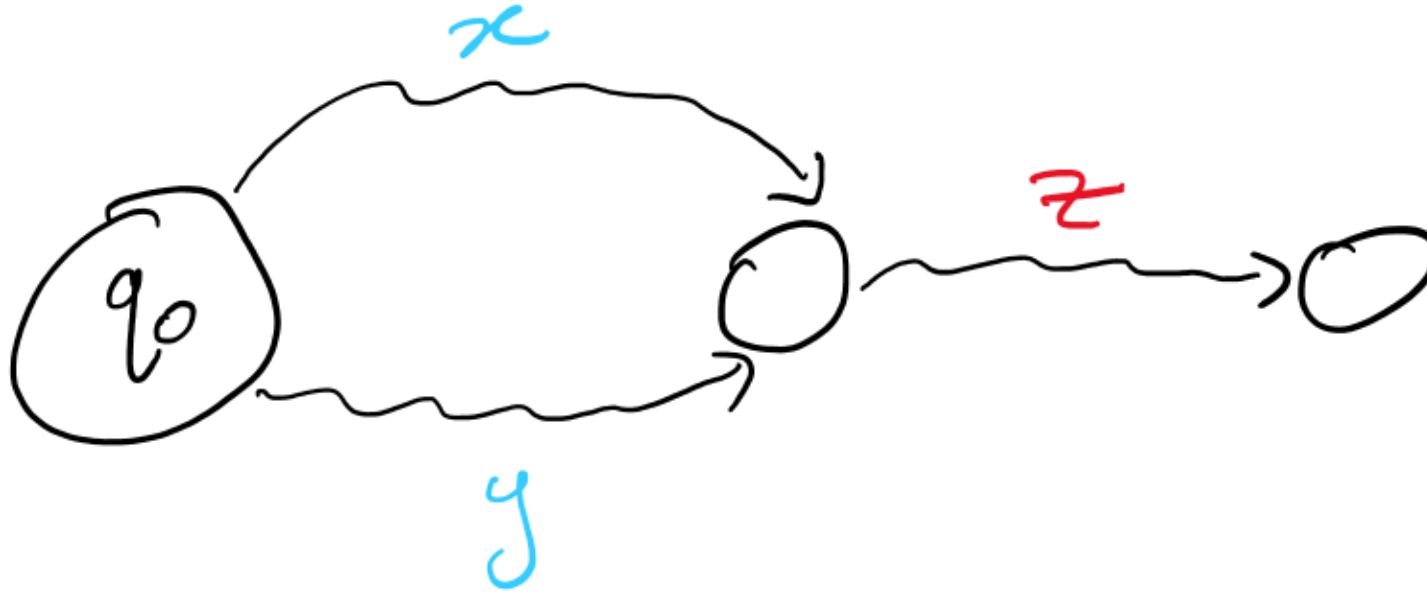
Observe:

If $w = xz$ and $u = yz$ are strings with same suffix z and $q(x) = q(y)$, then $q(xz) = q(yz)$.

Memorylessness Lemma

If $w = xz$ and $u = yz$ are strings with same suffix z and $q(x) = q(y)$, then $q(xz) = q(yz)$.

Consequently, M accepts both w and u , or rejects both of them



Memorylessness Lemma

If $w = xz$ and $u = yz$ are strings with same suffix z and $q(x) = q(y)$, then $q(xz) = q(yz)$.

Consequently, M accepts both w and u , or rejects both of them

Proof Sketch: Transition function depends only on current state and next symbol. Previous input symbols do not matter.

Application: If we can find $w = xz$ and $u = yz$ such that:

- exactly one of w, u is in L , and
- $q(x) = q(y)$,

then we can conclude that M makes a mistake on either w or u .

Thus, M does not recognize L 🎉

Distinguishable Pairs

- Let L be a language and x, y, z be strings
- z is a *distinguishing suffix* of x, y if **exactly one** of xz and yz is in L
- x and y is *distinguishable pair* if it has a distinguishing suffix

Example 1

- $L =$ set of even-length bit strings $= \{\epsilon, 00, 01, 10, 11, 0000, \dots\}$
- ϵ and 0 has distinguishing suffix 0 :
 - $\epsilon 0 = 0 \notin L$
 - $00 \in L$

Distinguishable Pairs

- Let L be a language and x, y, z be strings
- z is a *distinguishing suffix* of x, y if **exactly one** of xz and yz is in L
- x and y is *distinguishable pair* if it has a distinguishing suffix

Example 2

- $L =$ set of bit strings ending in 10 = $\{10, 010, 110, \dots\}$
- 1 and 0 has distinguishing suffix 0:
 - $00 \notin L$
 - $10 \in L$

Distinguishable Pairs

- Let L be a language and x, y, z be strings
- z is a *distinguishing suffix* of x, y if **exactly one** of xz and yz is in L
- x and y is *distinguishable pair* if it has a distinguishing suffix

Lemma. If x, y is distinguishable and M is a DFA such that $q(x) = q(y)$, then M does not recognize L .

Proof. Let z be distinguishing suffix. Memorylessness Lemma implies M accepts or rejects both xz and yz but exactly one of them is in L .

Fooling Sets

A set of strings F is a *fooling set* for language L if:

EVERY pair of strings $x, y \in F$ is distinguishable

Example:

- $L =$ set of bit strings ending in 10 = $\{10, 010, 110, \dots\}$
- $F = \{\epsilon, 1, 10\}$ is a fooling set
- ϵ and 1 has distinguishing suffix 0
- ϵ and 10 has distinguishing suffix ϵ
- 10 and 0 has distinguishing suffix ϵ

Fooling Sets

A set of strings F is a *fooling set* for language L if:

EVERY pair of strings $x, y \in F$ is distinguishable

Example:

- $L =$ set of bit strings ending in 10 = $\{10, 010, 110, \dots\}$
- $F = \{\epsilon, 1, 11\}$ is NOT a fooling set
- ϵ and 1 has distinguishing suffix 0
- ϵ and 11 has distinguishing suffix 0
- **BUT** 1 and 11 has **NO** distinguishing suffix

Fooling Set Lemma (Part 1)

If language L has a fooling set F with k strings, then it cannot be recognized by a DFA with less than k states.

Proof. Let M be a DFA with less than k states.

- There exists $x, y \in F$ such that $q(x) = q(y)$.
- Let z be distinguishing suffix of x, y
- Then, $q(xz) = q(yz)$ and so M accepts or rejects both xz and yz
- By definition, exactly one of xz and yz is in L
- So, M makes a mistake on one of them. QED

Fooling Set Lemma (Part 2)

If language L has a fooling set F_k with at least k strings for every positive integer k , then it cannot be recognized by any DFA.

Proof. Let M be a DFA and suppose it has k states.

- There exists $x, y \in F_{k+1}$ such that $q(x) = q(y)$.
- Let z be distinguishing suffix of x, y
- Then, $q(xz) = q(yz)$ and so M accepts or rejects both xz and yz
- By definition, exactly one of xz and yz is in L
- So, M makes a mistake on one of them. QED

Fooling Set Lemma: Application 1

- L = set of bit strings ending in 10 = $\{10, 010, 110, \dots\}$
- $F = \{\epsilon, 1, 10\}$ is a fooling set
- ϵ and 1 has distinguishing suffix 0
- ϵ and 10 has distinguishing suffix ϵ
- 10 and 0 has distinguishing suffix ϵ
- Thus, L can only be recognized by DFAs with at least 3 states

Fooling Set Lemma: Application 2

- $L = \{0^n 1^n \mid n \geq 0\} = \{\epsilon, 01, 0011, 000111, \dots\}$
- Warm up: Fooling set F_3 with 3 strings?
- $F_3 = \{1, 01, 001\}$ is a fooling set
- 1 and 01 has distinguishing suffix ϵ
- 1 and 001 has distinguishing suffix 1
- 01 and 001 has distinguishing suffix 1
- Thus, L can only be recognized by DFAs with at least 3 states
- How to construct a fooling set F_k with at least k strings for every k ?

0 / 1
01, 0011

1
01 — 001

Fooling Set Lemma: Application 2

- $L = \{0^n 1^n \mid n \geq 0\} = \{\epsilon, 01, 0011, 000111, \dots\}$
- $F_3 = \{1, 01, 001\}$ is a fooling set
- For every k , the set

$$F_k = \{0^i 1 \mid 0 \leq i \leq k - 1\} = \{1, 01, 001, \dots, 0^{k-1} 1\}$$

is a fooling set.

Proof.

For every pair of strings $x, y \in F_k$,

- $x = 0^i 1$ and $y = 0^j 1$ for $0 \leq j < i \leq k - 1$
- $z = 1^{i-1}$ is a distinguishing suffix
 - $xz = 0^i 1^i \in L$
 - $yz = 0^j 1^i \notin L$ since $j \neq i$

Summary: Key Concepts

- Memorylessness Lemma
- Distinguishable pairs and distinguishing suffixes
- Fooling sets
- Fooling Set Lemma